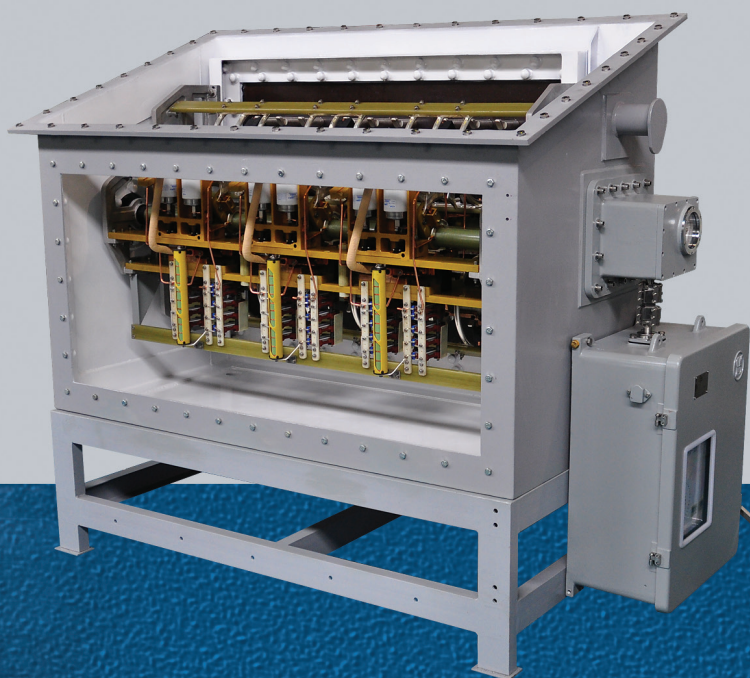




Type HWV Externally-mounted Vacuum OLTC

Shanghai Huaming Power Equipment Co.,Ltd.



Introduction of HWV externally-mounted OLTC



Type HWV externally-mounted vacuum OLTC is a new vacuum tap changer developed by Huaming. It is composed of diverter switch, tap selector, oil tank and motor drive unit. The diverter switch and tap selector of HWV is consolidated in one oil chamber. It provides the following features:

- 1.HWV is a vacuum switching tap changer with resistor transition;
- 2.As the arcing extinguished in the vacuum interrupter, no arcing occurs in the insulation oil. Thus, there is no need for oil filtration.
- 3.Since the insulation oil and insulating parts are free from contamination of carbons, the insulation reliability will greatly improved.
- 4.The current is carried by the shunt contacts at normal time.
The vacuum interrupters only carry the momentary current during switching process. So the tap changer has a high overload capacity.
- 5.The contacts of tap selector adopt the multi-point contacting type, which enhances their overload capacities greatly.
- 6.The compression spring of tap selector's contact is replaced by the composite leaf spring, which reduces the axial dimension effectively.
- 7.The diverter switch adopts new type energy accumulation mechanism.
- 8.Type HWV externally-mounted vacuum OLTC is completely separated from the transformer main tank so as to avoid contamination to the transformer oil. It is convenient for maintenance.

Type HWV OLTC is suitable for power and industrial transformers of Star or Delta connection. Its highest equipment voltage is 40.5 and 72.5kV; its three phases current is less than 1000A and rated frequency is 50 or 60 Hz.

The Structure Features of HWV

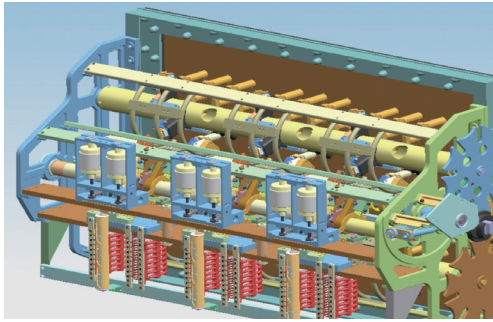


Fig.1

1. Good integral rigidity: diverter switch and tap selector are installed on the same bracket to ensure the accuracy of tap changer. (See Fig.1)

2. All transmission parts are fixed on one bracket to ensure the accuracy of their positions. (See Fig.2)

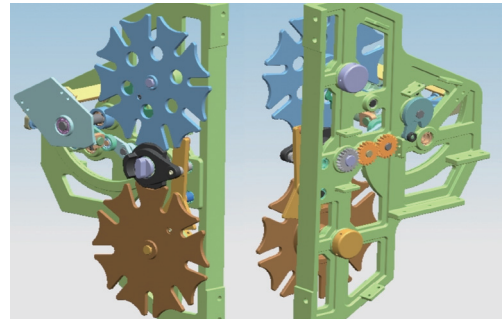


Fig.2

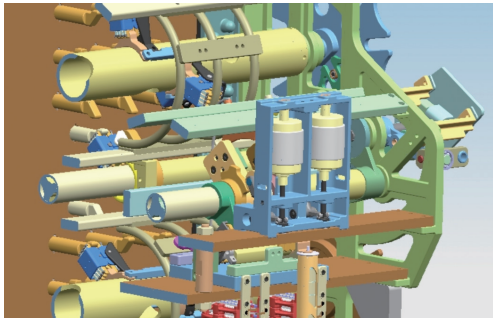


Fig.3

3. Odd and even shafts of tap selector are separated from each other and their contacts are directly connected on the tank wall. This can not only save the leads but also reduce the size of OLTC. (See Fig.3)

4. Use energy accumulation mechanism without claw clutch to ensure smooth operation. (See Fig.4)

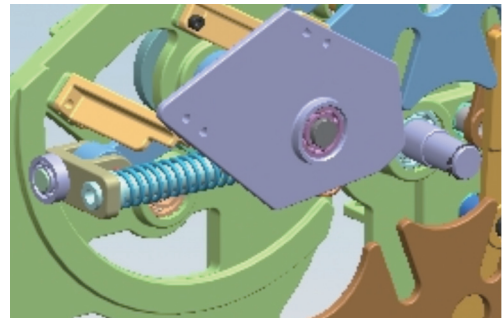


Fig.4

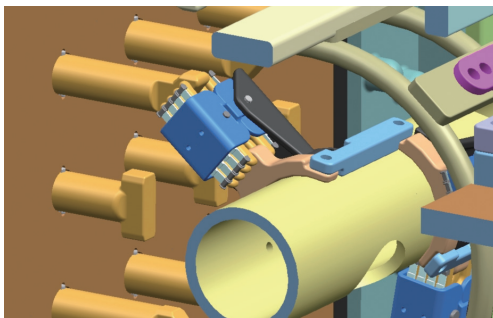


Fig.5

5. The contacts of tap selector adopt the multi-point contacting method, which enhances their overload capacities greatly. The compression spring of tap selector's contact is replaced by the composite leaf spring, which reduces the axial dimension effectively. (See Fig.5)

6. The current is carried by the shunt contacts at normal time. The vacuum interrupters only carry the momentary current during switching process. So the tap changer has a high overload capacity. (See Fig.6)

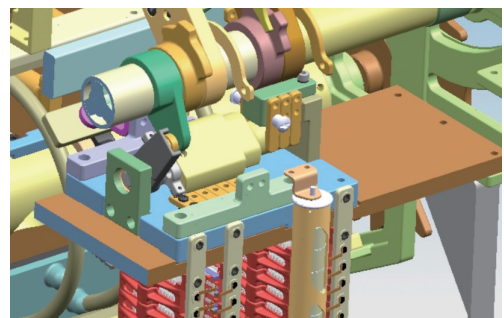


Fig.6

The Operating principles of HWV

HWV adopts double resistor transition principle. The switching sequence is shown in figure 7, the red line is the current.

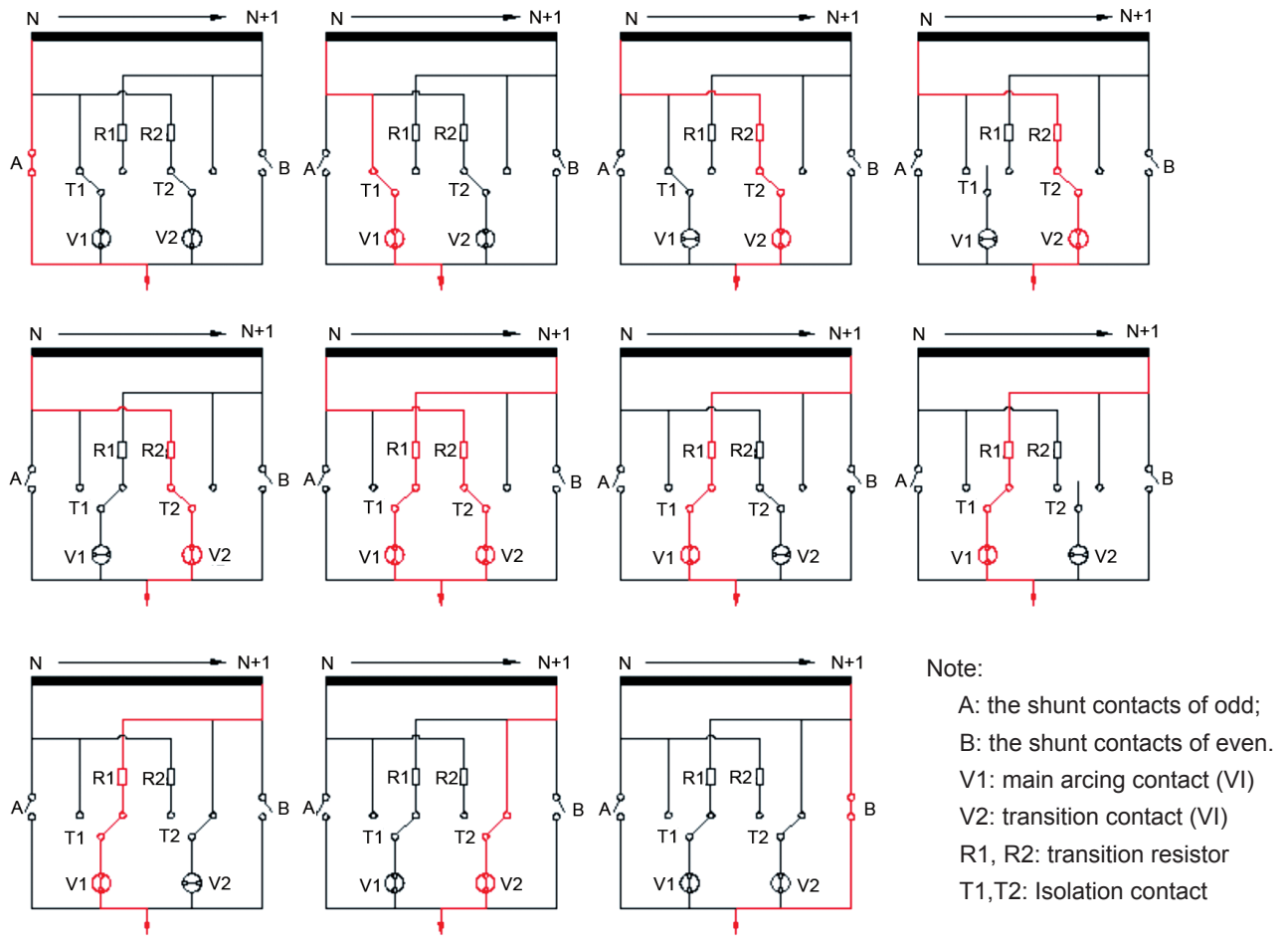


Fig.7

Introduction of HWV's Vacuum Interrupter

Vacuum interrupter, as the main component of HWV, decides the operating life of this tap changer.

- 1.HWV adopts the vacuum interrupter made by EATON;
- 2.Its mechanical life is 1,500,000 times , its electrical life is 300,000 times;
- 3.It is normally closed, the contact stroke is 5-5.5mm.

The Technical Data of HWV externally-mounted OLTC

Model		HWV 400	HWV 800	HWV 1000
Max. rated through current(A)		400	800	1000
Rated frequency (Hz)		50 or 60		
Phase & Connection		Y,D		
Max. rated step voltage (V)		3300		
Rated step capacity (kVA)		1200	2200	2600
Short-circuit current test(kA)	Thermal (3s)	8	8	12
	Dynamic (Peak)	20	20	30
Insulation to ground (KV)	Insulation Level			
	Highest voltage for equipment Um (KV)	17.5	40.5	72.5
	Rated separate source AC withstand voltage(50Hz ,1min)	45	90	140
	Rated lighting impulse withstand voltage (kV,1.2/50us)	150	250	350
Mechanical life		1,500,000 operations		
Contact life		300,000 operations		
Oil compartment of diverter switch	Service pressure	0.03MPa		
	Leakage test	0.08MPa 24h without leakage		
of diverter switch	Over pressure protection	Bursting cap bursts at 300±20%KPa		
	Protection relay	Oil flow speed set at 1.0m/s±10%		
Motor drive unit		SHM-D or CMA7		
OLTC Weight (kg)		800	1000	
Oil filling dm3		910	1140	

HWV Externally-mounted OLTC Insulation Level

Insulation distance		Withstand voltage	D connection	Y connection
a		BIL 1.2/50	265	265
		Power frequency 1Min	50	50
a1		BIL 1.2/50	90	90
		Power frequency 1Min	20	20
b	17.5kV	BIL 1.2/50	150	150
		Power frequency 1Min	45	45
	40.5kV	BIL 1.2/50	250	250
		Power frequency 1Min	90	90
	72.5kV	BIL 1.2/50	350	350
		Power frequency 1Min	140	140

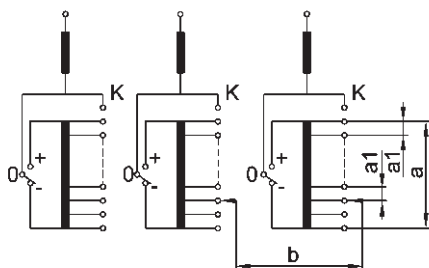
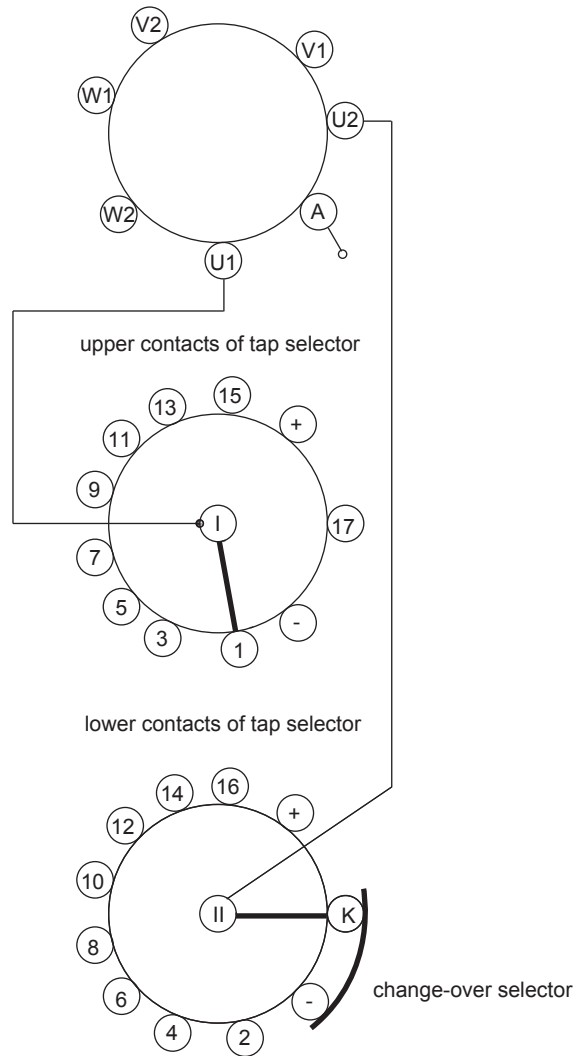
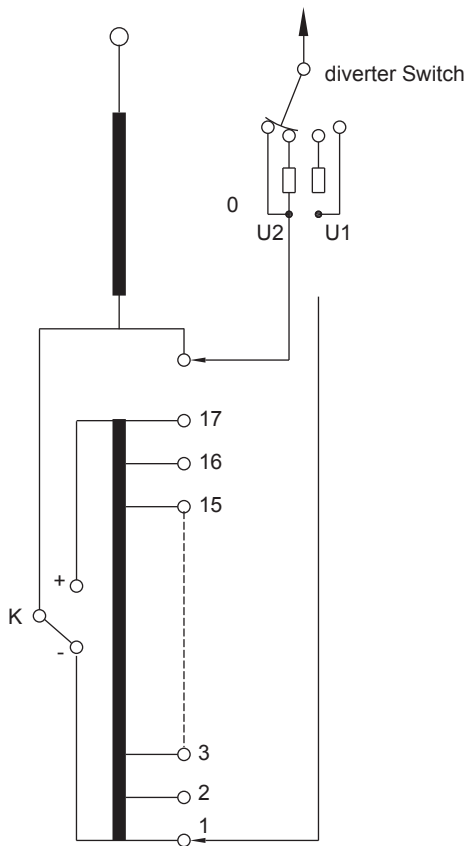


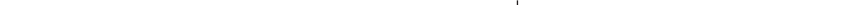
Fig.8

- a1: Between any two taps of the same phase;
- a: Between the beginning and the end taps of the same phase;
- b: Between any two taps of the different phase.

HWV Operating Position Table and Connection Diagram (18353W)



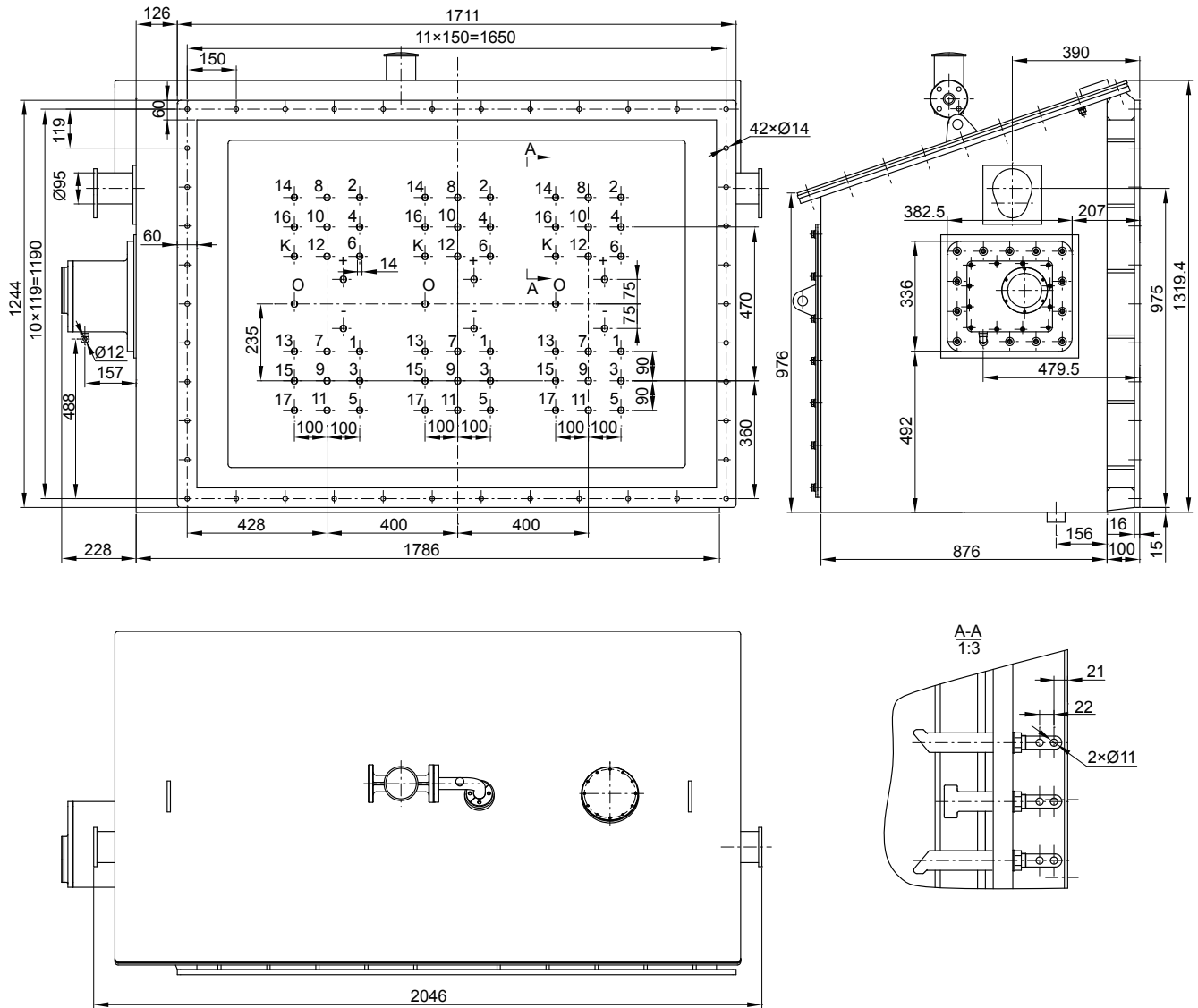
operating position number	35
different voltage number	33
set position ●	17b

change-over selector location																																			
tap selector contact position	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	k	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17
display position	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17a	17b	17c	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33



- drawing is shown at the set position

HWV Overall Dimensions



Note: The highest voltage for equipment is 72.5 kv.

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